

Three Random-Walk-Induced Message-Passing Paradigms on Hypergraphs

Adrián Ruiz Doblas

Abstract

This note places three vertex-level hypergraph message-passing paradigms (equal-nodes (EN), equal-edges (EE), and weighted-edges (WE)) within a common mathematical framework. It separates the transition kernel induced by each random walk from any additional structural descriptor supplied to the message function. The EE construction already developed in the submitted bachelor's thesis is recovered as one member of this family. The note then formalizes EN message passing, formalizes WE message passing, and proposes a WE-conditioned cardinality descriptor analogous to the existing EE descriptor. Five experiments, run under a shared protocol, test the resulting eight-variant ablation ladder on synthetic diagnostics and on the Cora and Citeseer co-citation benchmarks. This document extends the submitted work but does not modify the thesis or the presentation delivered at UAB.

1. Common notation and message-passing template

Let

$$H = (V, \mathcal{E})$$

be a finite undirected multi-hypergraph. Thus, $\mathcal{E}$ is regarded as an indexed family, and repeated hyperedges are counted with multiplicity. Hyperedges of cardinality one do not generate transitions between distinct vertices and are excluded from the propagation quantities below.

For each vertex $v \in V$, define

$$\mathcal{E}(v) := \{e \in \mathcal{E} : v \in e, |e| \geq 2\}, \quad d_H(v) := |\mathcal{E}(v)|.$$

For two distinct vertices $v, u \in V$, define

$$\mathcal{E}(v, u) := \{e \in \mathcal{E} : v, u \in e\}, \quad N_H(v) := \{u \in V \setminus \{v\} : \mathcal{E}(v, u) \neq \emptyset\}.$$

The pairwise co-occurrence multiplicity is

$$c_{vu} := |\mathcal{E}(v, u)|,$$

and the total number of possible non-self vertex choices across the hyperedges incident to v is

$$s_H(v) := \sum_{e \in \mathcal{E}(v)} (|e| - 1).$$

Equivalently,

$$s_H(v) = \sum_{u \in N_H(v)} c_{vu},$$

because every hyperedge $e \ni v$ contributes once to c_{vu} for each of the $|e| - 1$ vertices $u \in e \setminus \{v\}$.

We use the non-lazy convention: the walker always leaves its current vertex whenever the vertex is non-isolated. For each choice

$$X \in \{\text{EN}, \text{EE}, \text{WE}\},$$

let $P^X(v, u)$ denote the corresponding one-step transition probability. The basic random-walk-induced message-passing layer is

$$m_{v,X}^{(\ell+1)} := \sum_{u \in N_H(v)} P^X(v, u) \psi_X^{(\ell)}(h_v^{(\ell)}, h_u^{(\ell)}), \quad (1)$$

$$h_{v,X}^{(\ell+1)} := \phi_X^{(\ell)}(h_v^{(\ell)}, m_{v,X}^{(\ell+1)}). \quad (2)$$

Here $\psi_X^{(\ell)}$ is a shared learnable message function and $\phi_X^{(\ell)}$ is a shared learnable update function. If v is isolated, we set

$$m_{v,X}^{(\ell+1)} := \mathbf{0}.$$

Equations (1)-(2) are the message-passing template of Gilmer et al. (2017), specialized so that the aggregation weights are exactly the transition probabilities $P^X(v, u)$ of a random walk: the walk's only role in the template is to fix these weights. Additional descriptors may be supplied to the message function, but they should be distinguished from the transition kernel itself.

2. Equal-nodes message passing

The equal-nodes random walk selects one distinct neighbor of v uniformly. Its transition kernel is

$$P^{\text{EN}}(v, u) = \begin{cases} \frac{1}{|N_H(v)|}, & u \in N_H(v), \\ 0, & \text{otherwise.} \end{cases}$$

Consequently, the EN message-passing layer is

$$m_{v,\text{EN}}^{(\ell+1)} = \frac{1}{|N_H(v)|} \sum_{u \in N_H(v)} \psi_{\text{EN}}^{(\ell)}(h_v^{(\ell)}, h_u^{(\ell)}), \quad (3)$$

$$h_{v,\text{EN}}^{(\ell+1)} = \phi_{\text{EN}}^{(\ell)}(h_v^{(\ell)}, m_{v,\text{EN}}^{(\ell+1)}). \quad (4)$$

There is no canonical cardinality descriptor induced by the EN walk. EN chooses u directly from $N_H(v)$; it does not select an intermediate hyperedge. Therefore, unlike EE and WE, the walk itself does not define a conditional distribution over the hyperedges that could have produced the transition $v \rightarrow u$. One may attach an independently constructed incidence descriptor to an EN-weighted layer, but that descriptor would not arise from the EN random walk and the resulting architecture would no longer be a pure EN construction.

The kernel P^{EN} is exactly the transition kernel of the ordinary random walk on the unweighted clique expansion. Hence it only retains whether two vertices are adjacent in the hypergraph, i.e., $\mathcal{E}(v, u) \neq \emptyset$, and it ignores both the number and the cardinalities of the hyperedges shared by the pair.

3. Equal-edges message passing

The equal-edges random walk first selects uniformly one hyperedge $e \in \mathcal{E}(v)$, and then selects uniformly one vertex $u \in e \setminus \{v\}$. Thus,

$$\mathbb{P}(e \mid v) = \frac{1}{d_H(v)}, \quad \mathbb{P}(u \mid v, e) = \frac{1}{|e| - 1}.$$

Since the EE walk is the two-step process just described, its one-step transition probability $P^{\text{EE}}(v, u)$ is, by the law of total probability, the marginal probability $\mathbb{P}(u \mid v)$ of this process (the sum restricted to $\mathcal{E}(v, u)$ since $\mathbb{P}(u \mid v, e) = 0$ for $e \notin \mathcal{E}(v, u)$):

$$P^{\text{EE}}(v, u) = \mathbb{P}(u \mid v) = \sum_{e \in \mathcal{E}(v, u)} \mathbb{P}(e \mid v) \mathbb{P}(u \mid v, e) = \frac{1}{d_H(v)} \sum_{e \in \mathcal{E}(v, u)} \frac{1}{|e| - 1}.$$

Define the unnormalized equal-edges pair weight

$$Z_{vu}^{\text{EE}} := \sum_{e \in \mathcal{E}(v, u)} \frac{1}{|e| - 1}.$$

Then

$$P^{\text{EE}}(v, u) = \frac{Z_{vu}^{\text{EE}}}{d_H(v)}.$$

The basic EE message-passing layer is

$$m_{v, \text{EE}}^{(\ell+1)} = \sum_{u \in N_H(v)} P^{\text{EE}}(v, u) \psi_{\text{EE}}^{(\ell)}(h_v^{(\ell)}, h_u^{(\ell)}), \quad (5)$$

$$h_{v, \text{EE}}^{(\ell+1)} = \phi_{\text{EE}}^{(\ell)}(h_v^{(\ell)}, m_{v, \text{EE}}^{(\ell+1)}). \quad (6)$$

3.1. The EE cardinality descriptor

To retain more information than the scalar transition probability alone, fix

$$K := \max \{|e| : e \in \mathcal{E}, |e| \geq 2\}$$

¹. For every cardinality $r \geq 2$, let

$$\mathbf{c}_r \in \{0, 1\}^{K-1}$$

¹With this canonical choice, cardinalities $2, \dots, K$ each receive their own bin, so no distinct observed cardinality is folded into another; a smaller K would fold distinct cardinalities together, while any $K \geq \max_e |e|$ preserves this exactness, only adding unpopulated coordinates. χ_{vu}^{EE} alone is a histogram and so does not identify the multiset of shared cardinalities when several hyperedges land in the same bin; recovering that multiset from the pair information requires $(Z_{vu}^{\text{EE}}, \chi_{vu}^{\text{EE}})$ jointly, as in Proposition 9.2.

be the one-hot vector whose nonzero entry is at position

$$b(r) := r - 1.$$

Under the canonical choice $K := \max_e |e|$, cardinalities $2, \dots, K$ each occupy a separate bin, so every observed cardinality receives its own coordinate.

For every adjacent ordered pair (v, u) , define

$$\chi_{vu}^{\text{EE}} := \frac{1}{Z_{vu}^{\text{EE}}} \sum_{e \in \mathcal{E}(v, u)} \frac{1}{|e| - 1} \mathbf{c}_{|e|}.$$

The enriched EE layer is

$$m_{v, \text{EE}+}^{(\ell+1)} = \sum_{u \in N_H(v)} P^{\text{EE}}(v, u) \psi_{\text{EE}+}^{(\ell)} \left(h_v^{(\ell)}, h_u^{(\ell)}, \chi_{vu}^{\text{EE}} \right), \quad (7)$$

$$h_{v, \text{EE}+}^{(\ell+1)} = \phi_{\text{EE}+}^{(\ell)} \left(h_v^{(\ell)}, m_{v, \text{EE}+}^{(\ell+1)} \right). \quad (8)$$

The descriptor has a direct probabilistic interpretation. By Bayes' rule, the conditional probability that the transition $v \rightarrow u$ was produced through a particular shared hyperedge $e \in \mathcal{E}(v, u)$ is

$$\mathbb{P}(e \mid v, u) = \frac{\mathbb{P}(e \mid v) \mathbb{P}(u \mid v, e)}{\mathbb{P}(u \mid v)} = \frac{1/(|e| - 1)}{Z_{vu}^{\text{EE}}}.$$

Therefore,

$$\chi_{vu}^{\text{EE}} = \sum_{e \in \mathcal{E}(v, u)} \mathbb{P}(e \mid v, u) \mathbf{c}_{|e|},$$

that is, χ_{vu}^{EE} is exactly the binned distribution of $|e|$ under $\mathbb{P}(\cdot \mid v, u)$, matching the definition of χ_{vu}^{EE} given above.

Small hyperedges receive more conditional mass because, after a hyperedge is selected, their probability is distributed among fewer possible destinations.

The scalar kernel P^{EE} is already cardinality-sensitive, since every shared hyperedge contributes $(|e| - 1)^{-1}$. However, it compresses all these contributions into one scalar and generally does not identify the multiset or distribution of shared hyperedge cardinalities. The role of χ_{vu}^{EE} is precisely to expose part of that cardinality pattern to the learnable message function.

Equivalently, P^{EE} is the random-walk kernel on a weighted clique expansion with pair weight

$$W_{vu}^{\text{EE}} := Z_{vu}^{\text{EE}} = \sum_{e \in \mathcal{E}(v, u)} \frac{1}{|e| - 1}.$$

The enriched construction may therefore be represented as message passing on this weighted pairwise support with χ_{vu}^{EE} as an additional edge attribute. The quantities remain hypergraph-derived, but the resulting one-step operator is pairwise once these quantities have been precomputed.

4. Weighted-edges message passing

The weighted-edges random walk first selects an incident hyperedge with probability proportional to the number of possible destinations that it contains. Under the non-lazy convention,

$$\mathbb{P}(e \mid v) = \frac{|e| - 1}{s_H(v)}.$$

It then selects a vertex $u \in e \setminus \{v\}$ uniformly:

$$\mathbb{P}(u \mid v, e) = \frac{1}{|e| - 1}.$$

Since the WE walk is the two-step process just described, its one-step transition probability $P^{\text{WE}}(v, u)$ is, by the law of total probability, the marginal probability $\mathbb{P}(u \mid v)$ of this process. Hence, the factors $|e| - 1$ cancel:

$$P^{\text{WE}}(v, u) = \mathbb{P}(u \mid v) = \sum_{e \in \mathcal{E}(v, u)} \mathbb{P}(e \mid v) \mathbb{P}(u \mid v, e) = \sum_{e \in \mathcal{E}(v, u)} \frac{|e| - 1}{s_H(v)} \frac{1}{|e| - 1} \quad (9)$$

$$= \frac{|\mathcal{E}(v, u)|}{s_H(v)} = \frac{c_{vu}}{s_H(v)}. \quad (10)$$

This is the non-lazy WE kernel of the ORCHID framework Coupette et al. (2023). The basic WE message-passing layer is

$$m_{v, \text{WE}}^{(\ell+1)} = \sum_{u \in N_H(v)} P^{\text{WE}}(v, u) \psi_{\text{WE}}^{(\ell)}(h_v^{(\ell)}, h_u^{(\ell)}), \quad (11)$$

$$h_{v, \text{WE}}^{(\ell+1)} = \phi_{\text{WE}}^{(\ell)}(h_v^{(\ell)}, m_{v, \text{WE}}^{(\ell+1)}). \quad (12)$$

Thus, WE assigns more transition mass to a neighbor when that neighbor co-occurs with v in more hyperedges. Unlike EE, the final pairwise weight does not depend on the cardinality of each shared hyperedge individually: each shared hyperedge contributes exactly one unit to c_{vu} .

4.1. A natural WE cardinality descriptor

For WE, define

$$\chi_{vu}^{\text{WE}} := \frac{1}{c_{vu}} \sum_{e \in \mathcal{E}(v, u)} \mathbf{c}_{|e|}.$$

The corresponding enriched WE layer is

$$m_{v, \text{WE}+}^{(\ell+1)} = \sum_{u \in N_H(v)} P^{\text{WE}}(v, u) \psi_{\text{WE}+}^{(\ell)}(h_v^{(\ell)}, h_u^{(\ell)}, \chi_{vu}^{\text{WE}}), \quad (13)$$

$$h_{v, \text{WE}+}^{(\ell+1)} = \phi_{\text{WE}+}^{(\ell)}(h_v^{(\ell)}, m_{v, \text{WE}+}^{(\ell+1)}). \quad (14)$$

The descriptor has a direct probabilistic interpretation. For a fixed transition $v \rightarrow u$, the joint probability that the WE walk reaches u through a particular shared hyperedge $e \in \mathcal{E}(v, u)$ is

$$\mathbb{P}(e, u \mid v) = \mathbb{P}(e \mid v) \mathbb{P}(u \mid v, e) = \frac{|e| - 1}{s_H(v)} \cdot \frac{1}{|e| - 1} = \frac{1}{s_H(v)},$$

where we have used the chain rule. By Bayes' rule,

$$\mathbb{P}(e \mid v, u) = \frac{\mathbb{P}(e, u \mid v)}{\mathbb{P}(u \mid v)} = \frac{1/s_H(v)}{c_{vu}/s_H(v)} = \frac{1}{c_{vu}},$$

constant in e : conditional on the event that the WE walk moves from v to u , the shared hyperedge is uniformly distributed over $\mathcal{E}(v, u)$. Therefore,

$$\chi_{vu}^{\text{WE}} = \sum_{e \in \mathcal{E}(v, u)} \mathbb{P}(e \mid v, u) \mathbf{c}_{|e|},$$

that is, χ_{vu}^{WE} is exactly the binned distribution of $|e|$ under $\mathbb{P}(\cdot \mid v, u)$, matching the definition of χ_{vu}^{WE} given above. Unlike χ_{vu}^{EE} , this histogram weights all shared hyperedges equally. This descriptor is not part of the original definition of the WE random walk, but a new extension obtained by applying the same conditional-cardinality principle used in the submitted EE-based construction.

The scalar kernel P^{WE} , by contrast, discards cardinality information entirely: as shown above, every shared hyperedge contributes exactly one unit to c_{vu} regardless of its size. The role of χ_{vu}^{WE} is precisely to expose that discarded cardinality pattern to the learnable message function.

Example 4.1. Suppose that v and u share exactly two hyperedges,

$$\mathcal{E}(v, u) = \{e_1, e_2\}, \quad |e_1| = 3, \quad |e_2| = 5.$$

Suppose further that no hyperedge of H exceeds cardinality 5, so that $K := \max_e |e| = 5$ is realized by e_2 itself. The four coordinates then represent cardinalities 2, 3, 4, 5 exactly, with no cardinality folded into an overflow bin. Hence,

$$\mathbf{c}_3 = (0, 1, 0, 0), \quad \mathbf{c}_5 = (0, 0, 0, 1).$$

For the EE descriptor,

$$Z_{vu}^{\text{EE}} = \frac{1}{3-1} + \frac{1}{5-1} = \frac{3}{4},$$

and therefore

$$\chi_{vu}^{\text{EE}} = \frac{1}{3/4} \left(\frac{1}{2} \mathbf{c}_3 + \frac{1}{4} \mathbf{c}_5 \right) = \frac{2}{3} \mathbf{c}_3 + \frac{1}{3} \mathbf{c}_5 = \left(0, \frac{2}{3}, 0, \frac{1}{3} \right).$$

For the WE descriptor, both shared hyperedges receive the same weight:

$$\chi_{vu}^{\text{WE}} = \frac{1}{2} (\mathbf{c}_3 + \mathbf{c}_5) = \left(0, \frac{1}{2}, 0, \frac{1}{2} \right).$$

Thus, EE gives more weight to the smaller hyperedge, whereas WE weights both shared hyperedges equally.

Equivalently, P^{WE} is the random-walk kernel on a weighted clique expansion with pair weight

$$W_{vu}^{\text{WE}} := c_{vu}.$$

The enriched variant adds χ_{vu}^{WE} as a pairwise edge attribute.

Remark 4.2 (A common principle behind both descriptors). *Both cardinality descriptors instantiate the same construction: the posterior expectation of the shared hyperedge’s cardinality, conditional on the transition itself,*

$$\chi_{vu} := \mathbb{E} \left[\mathbf{c}_{|e|} \mid v \rightarrow u \right] = \sum_{e \in \mathcal{E}(v,u)} \mathbb{P}(e \mid v, u) \mathbf{c}_{|e|}.$$

Substituting the posterior $\mathbb{P}(e \mid v, u)$ derived above for each walk recovers χ_{vu}^{EE} and χ_{vu}^{WE} exactly, so the two descriptors differ only in which conditional law each random walk induces, not in the principle used to build a descriptor from it.

5. Information retained by each paradigm

The kernels differ in the structural information used to weight the same pairwise support

$$\{(v, u) : u \in N_H(v)\}.$$

Table 1 summarizes the distinction, and Figure 1 draws it on one example.

Table 1: pairwise representations and retained structural information.

Paradigm	Equivalent pairwise object	Information retained by the propagation kernel
EN	Unweighted clique expansion	Whether v and u share at least one hyperedge. Multiplicity and hyperedge cardinality are discarded.
EE	Weighted clique expansion with $W_{vu}^{\text{EE}} = \sum_{e \in \mathcal{E}(v,u)} (e - 1)^{-1}$	Co-occurrence multiplicity and cardinality are combined into one reciprocal-cardinality-weighted scalar.
WE	Weighted clique expansion with $W_{vu}^{\text{WE}} = c_{vu}$	The number of hyperedges shared by v and u ; the individual cardinalities of those hyperedges do not affect the scalar pair weight.
EE+ χ^{EE}	Weighted pairwise graph with edge attributes	The EE scalar weight together with a reciprocal-cardinality-weighted histogram of shared hyperedge sizes.
WE+ χ^{WE}	Weighted pairwise graph with edge attributes	The co-occurrence count together with an equally weighted histogram of shared hyperedge sizes.

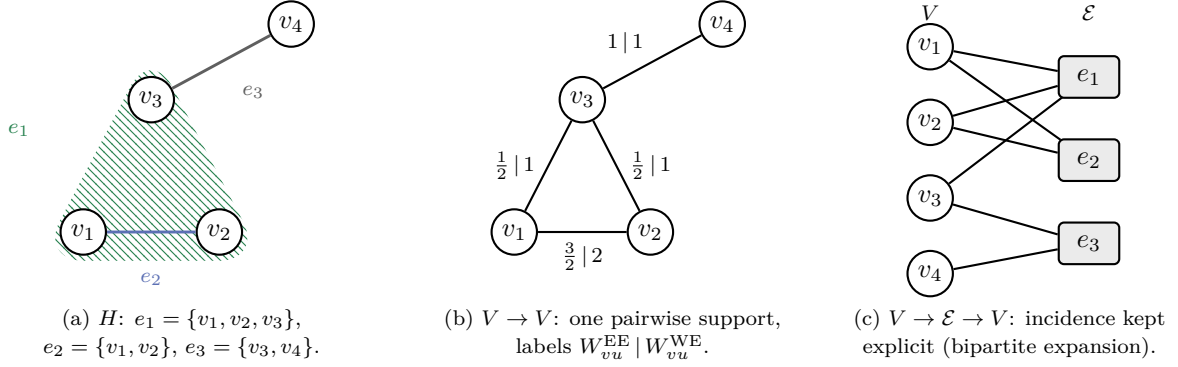


Figure 1: the same hypergraph under the two propagation regimes. All three kernels of this note act on the single pairwise support of (b): P^{EN} uses it unweighted, P^{EE} and P^{WE} weight it by W^{EE} and W^{WE} respectively, and the Pattern layers add χ_{vu} as an edge attribute. Panel (c) is the alternative of Section 7, where hyperedges receive their own representations instead.

Proposition 5.1 (Row-stochasticity). *For every non-isolated vertex $v \in V$, each of $P^{\text{EN}}(v, \cdot)$, $P^{\text{EE}}(v, \cdot)$, $P^{\text{WE}}(v, \cdot)$ is a probability distribution over $N_H(v)$.*

Proof. For EN this is immediate, since $P^{\text{EN}}(v, u) = 1/|N_H(v)|$ is uniform over $N_H(v)$. For EE, substituting the definition of $P^{\text{EE}}(v, u)$ and exchanging the order of summation ($u \in N_H(v)$ paired with $e \in \mathcal{E}(v, u)$ ranges over the same pairs (u, e) as $e \in \mathcal{E}(v)$ paired with $u \in e \setminus \{v\}$),

$$\begin{aligned} \sum_{u \in N_H(v)} P^{\text{EE}}(v, u) &= \sum_{u \in N_H(v)} \frac{1}{d_H(v)} \sum_{e \in \mathcal{E}(v, u)} \frac{1}{|e| - 1} \\ &= \frac{1}{d_H(v)} \sum_{e \in \mathcal{E}(v)} \sum_{u \in e \setminus \{v\}} \frac{1}{|e| - 1}. \end{aligned}$$

For fixed e , the set $e \setminus \{v\}$ has exactly $|e| - 1$ elements, so the inner sum adds the constant $1/(|e| - 1)$ over $|e| - 1$ terms:

$$\begin{aligned} \sum_{u \in N_H(v)} P^{\text{EE}}(v, u) &= \frac{1}{d_H(v)} \sum_{e \in \mathcal{E}(v)} (|e| - 1) \cdot \frac{1}{|e| - 1} \\ &= \frac{1}{d_H(v)} \sum_{e \in \mathcal{E}(v)} 1. \end{aligned}$$

Since $d_H(v) := |\mathcal{E}(v)|$, the remaining sum has exactly $d_H(v)$ terms, each equal to 1:

$$\sum_{u \in N_H(v)} P^{\text{EE}}(v, u) = \frac{1}{d_H(v)} \cdot d_H(v) = 1.$$

For WE, it is more straightforward: just by using the definitions of P^{WE} and s_H , we have

$$\sum_{u \in N_H(v)} P^{\text{WE}}(v, u) = \frac{\sum_{u \in N_H(v)} c_{vu}}{s_H(v)} = 1.$$

□

Remark 5.2 (Equivariance under relabeling). *All three constructions are also equivariant under relabeling of the vertices, provided that the same message and update functions are shared across vertices: vertex permutations preserve adjacency, hyperedge cardinalities, co-occurrence multiplicities, and all the sums defining the kernels and descriptors.*

Definition 5.3 (Uniform, regular, and co-occurrent hypergraphs). *A hypergraph $H = (V, \mathcal{E})$ is called r -uniform if*

$$|e| = r \quad \text{for every } e \in \mathcal{E},$$

k -regular if

$$d_H(v) = k \quad \text{for every } v \in V,$$

and c -cooccurrent if every pair of adjacent vertices belongs to exactly c common hyperedges, that is,

$$c_{vu} = |\mathcal{E}(v, u)| = c \quad \text{for every } u \in N_H(v).$$

Proposition 5.4 (Coincidence in homogeneous regimes). *Let H be r -uniform and c -cooccurrent. Then the three kernels coincide: for every adjacent ordered pair (v, u) ,*

$$P^{\text{EN}}(v, u) = P^{\text{EE}}(v, u) = P^{\text{WE}}(v, u) = \frac{c}{s_H(v)}.$$

If H is in addition k -regular, then $s_H(v) = k(r - 1)$ for every vertex and the common value takes the vertex-independent form $c/(k(r - 1))$ Coupette et al. (2023). Simple graphs are the special case $r = 2$, $c = 1$, for which the common value is $1/d_H(v)$; regularity is not needed there either.

Proof. Assume c -cooccurrence. Then $s_H(v) = \sum_{u \in N_H(v)} c_{vu} = c |N_H(v)|$, so

$$P^{\text{EN}}(v, u) = \frac{1}{|N_H(v)|} = \frac{c}{s_H(v)}, \quad P^{\text{WE}}(v, u) = \frac{c_{vu}}{s_H(v)} = \frac{c}{s_H(v)}.$$

Assume now r -uniformity. Then $Z_{vu}^{\text{EE}} = c_{vu}/(r - 1)$ and $s_H(v) = (r - 1)d_H(v)$, so

$$P^{\text{EE}}(v, u) = \frac{Z_{vu}^{\text{EE}}}{d_H(v)} = \frac{c_{vu}}{(r - 1)d_H(v)} = \frac{c_{vu}}{s_H(v)} = \frac{c}{s_H(v)}.$$

Under k -regularity, $s_H(v) = \sum_{e \in \mathcal{E}(v)} (r - 1) = k(r - 1)$. □

Remark 5.5 (Both hypotheses are needed, and regularity is not). *Neither hypothesis of Proposition 5.4 can be dropped. On the 3-uniform hypergraph $\{0, 1, 2\}, \{0, 1, 3\}$, which is not c -cooccurrent, $P^{\text{EN}}(0, 1) = 1/3$ while $P^{\text{EE}}(0, 1) = P^{\text{WE}}(0, 1) = 1/2$. On the 1-cooccurrent hypergraph $\{0, 1\}, \{0, 2, 3\}$, which is not uniform, $P^{\text{EE}}(0, 1) = 1/2$ while $P^{\text{EN}}(0, 1) = P^{\text{WE}}(0, 1) = 1/3$. Regularity, by contrast, is not needed: on $\{0, 1, 2\}, \{3, 4, 5\}, \{5, 6, 7\}$, which is 3-uniform and 1-cooccurrent but not regular, the three kernels still agree at every vertex. The proof above isolates exactly which hypothesis each equality uses: $\text{EN} = \text{WE}$ needs only c -cooccurrence, and $\text{EE} = \text{WE}$ needs only r -uniformity. Figure 2 draws the first two hypergraphs with all three rows written out.*

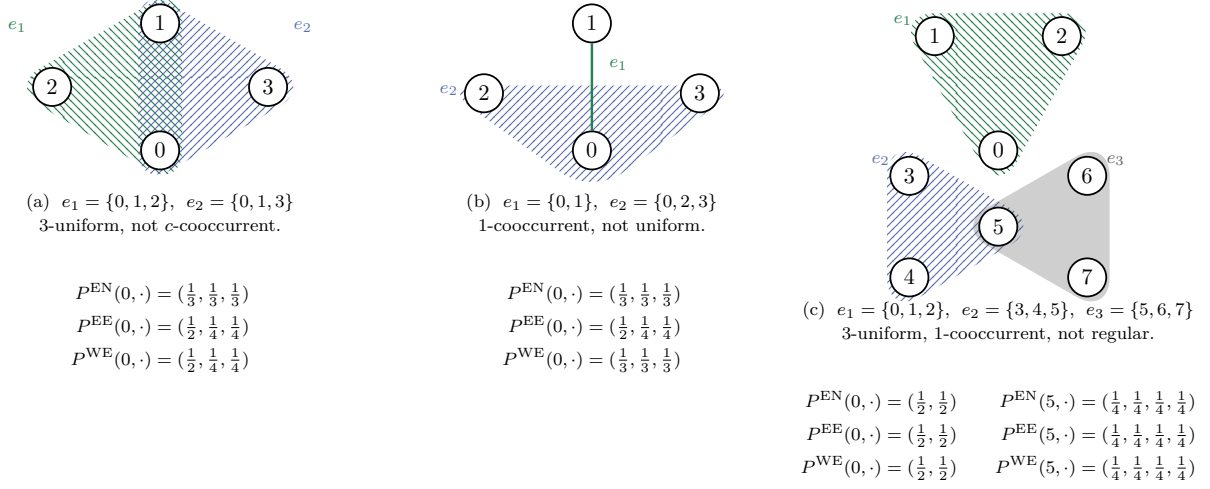


Figure 2: (a) and (b) are the two counterexamples of the remark above, with rows ordered (1, 2, 3): each drops one hypothesis of Proposition 5.4, and the coincidence breaks. Uniformity alone still gives $P^{\text{EE}} = P^{\text{WE}}$ in (a), cooccurrence alone still gives $P^{\text{EN}} = P^{\text{WE}}$ in (b), and the P^{EE} row is the same on both sides while its companion changes. (c) is not a counterexample: it drops regularity only, and the three kernels still agree at every vertex, though the common value depends on the vertex, $1/2$ at vertex 0 against $1/4$ at vertex 5.

Remark 5.6 (Multi-edges). *Since $\mathcal{E}$ is an indexed family, a graph with parallel edges is 2-uniform but need not be 1-cooccurrent, and then the kernels differ. On $\{0, 1\}$, $\{0, 1\}$, $\{0, 2\}$ one has $P^{\text{EN}}(0, \cdot) = (1/2, 1/2)$ but $P^{\text{EE}}(0, \cdot) = P^{\text{WE}}(0, \cdot) = (2/3, 1/3)$. The coincidence for ordinary graphs is therefore a statement about simple graphs.*

Outside the homogeneous regime of Proposition 5.4, the three kernels may assign genuinely different transition distributions to the same vertex.

Remark 5.7 (Role of the descriptors). *The comparison also clarifies the role of the descriptors. The transition kernel determines how much the message from u contributes to the aggregation at v . A descriptor such as χ_{vu}^{EE} or χ_{vu}^{WE} modifies what message is computed from that ordered pair. The descriptor is deterministic and parameter-free; all learning remains inside the message and update functions.*

6. What is added beyond the submitted thesis and presentation

The submitted work already contained the following elements:

1. The non-lazy equal-edges transition kernel P^{EE} .
2. The cardinality-pattern descriptor denoted there by χ_{vu} , which is written here as χ_{vu}^{EE} to distinguish it from the new WE analogue.
3. The complete EE-based EO-Pattern layer, corresponding to Equation (7).
4. An ablation using uniform aggregation over $N_H(v)$, although it was presented as a baseline rather than as a separately formalized EN paradigm.

The present note adds the following material without changing the submitted documents:

1. **A common formal framework.** EN, EE, and WE are written as three instances of the same message/aggregate/update template, making the distinction between the random-walk kernel and the message function explicit.
2. **A formal EN layer.** The previous uniform aggregation baseline is identified mathematically as equal-nodes message passing, since its weights are exactly $1/|N_H(v)|$. Following Marzieh Eidi’s recommendation in personal communication (and all the logical reasoning developed above), no χ -descriptor is attached to the pure EN formulation.
3. **A formal WE layer.** The weighted-edges random walk is converted into a message-passing operator through the kernel

$$P^{\text{WE}}(v, u) = \frac{c_{vu}}{s_H(v)}.$$

This yields a third propagation paradigm that was not formalized as a layer in the submitted thesis.

4. **A new WE-conditioned cardinality descriptor.** The quantity

$$\chi_{vu}^{\text{WE}} = \frac{1}{c_{vu}} \sum_{e \in \mathcal{E}(v, u)} \mathbf{c}_{|e|}$$

is proposed as the natural WE analogue of the existing EE descriptor. It is new: it is not part of the definition of the WE random walk in ORCHID and was not contained in the submitted work.

5. **A probabilistic interpretation of both descriptors.** The existing χ_{vu}^{EE} and the new χ_{vu}^{WE} are interpreted uniformly as conditional distributions of hyperedge cardinality given that the corresponding random walk performs the transition $v \rightarrow u$. EE yields reciprocal-cardinality weighting, whereas WE yields uniform weighting over the shared hyperedges.
6. **A precise pairwise comparison.** The note identifies the weighted or unweighted clique graph on which each scalar kernel acts and states exactly what information each kernel preserves or compresses. This separates the claim that the quantities are derived from a hypergraph from the separate fact that the resulting one-step operators can be represented on pairwise attributed graphs after preprocessing.
7. **Row-stochasticity and coincidence conditions, generalized.** The submitted thesis proved row-stochasticity only for P^{EE} and a collapse result only under the narrower hypothesis of a k -uniform, linear ($c = 1$ -cooccurrent) hypergraph, comparing P^{EE} implicitly against uniform aggregation alone. Proposition 5.1 here extends row-stochasticity to all three kernels. Proposition 5.4 generalizes the collapse result to general c -cooccurrence and to all three paradigms simultaneously, and additionally isolates exactly which hypothesis each pairwise equality needs, showing in particular that regularity is not required for any of them.

Accordingly, the new conceptual organization is

EN-MP,	EE-MP,	WE-MP
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for the three pure random-walk-induced layers, together with

$$\boxed{\text{EE-Pattern-MP}, \quad \text{WE-Pattern-MP}}$$

for the two cardinality-conditioned extensions. The submitted EO-Pattern model is precisely the first of these two enriched constructions.

7. Relation to hypergraph neural networks

7.1. Direct vertex-to-vertex propagation

The EN, EE, and WE constructions use direct vertex-to-vertex propagation:

$$V \longrightarrow V.$$

After P^X and, when applicable, χ^X have been computed from the hypergraph, all learnable hidden representations remain attached to vertices. A layer aggregates messages of the form

$$u \longrightarrow v$$

directly over $u \in N_H(v)$.

Consequently, each construction can be represented as message passing on an unweighted, weighted, or edge-attributed graph on the vertex set V , as in Figure 1(b). The hypergraph determines this pairwise representation during preprocessing, but hyperedges do not receive their own hidden representations during the neural computation.

7.2. Vertex-to-hyperedge-to-vertex propagation

Many hypergraph neural networks instead propagate through the incidence structure of Figure 1(c) using two stages:

$$V \longrightarrow \mathcal{E} \longrightarrow V.$$

A generic layer first constructs a representation for every hyperedge,

$$z_e^{(\ell)} = \rho^{(\ell)} \left(\{h_u^{(\ell)} : u \in e\} \right),$$

and then updates every vertex using its incident hyperedges,

$$h_v^{(\ell+1)} = \eta^{(\ell)} \left(h_v^{(\ell)}, \{z_e^{(\ell)} : e \in \mathcal{E}(v)\} \right),$$

where the set aggregations are permutation-invariant. Architectures such as AllSet follow this general two-stage principle Chien et al. (2022).

This approach does not first collapse the hypergraph into pairwise interactions. It can therefore use each hyperedge as an explicit intermediate object and can naturally incorporate hyperedge features or learned hyperedge representations.

The present $V \rightarrow V$ construction is simpler and keeps only vertex states, but it retains only the hypergraph information encoded by P^X and χ^X ; it can nonetheless be represented within the relational message-passing framework of Taha et al. (2025), by taking the vertex set as the entity set and encoding each kernel P^X as a shift operator on a binary adjacency relation, with an auxiliary pair entity added for the Pattern variants; Appendix A formalizes this representation. The $V \rightarrow \mathcal{E} \rightarrow V$ approach preserves the incidence grouping more explicitly, at the cost of an additional aggregation stage and a more complex architecture.

8. Precise form of the learnable maps

Assume that

$$h_v^{(\ell)} \in \mathbb{R}^{d_\ell}$$

is the representation of vertex v at layer ℓ . For the pure EN, EE, and WE layers, the message function has the form

$$\psi_X^{(\ell)} : \mathbb{R}^{d_\ell} \times \mathbb{R}^{d_\ell} \longrightarrow \mathbb{R}^{p_\ell}, \quad X \in \{\text{EN}, \text{EE}, \text{WE}\}.$$

It maps the representations of the target and source vertices to a message vector:

$$\psi_X^{(\ell)}(h_v^{(\ell)}, h_u^{(\ell)}) \in \mathbb{R}^{p_\ell}.$$

For the EE-Pattern and WE-Pattern variants, the descriptor satisfies

$$\chi_{vu}^X \in \mathbb{R}^{K-1}, \quad X \in \{\text{EE}, \text{WE}\},$$

and the message function becomes

$$\psi_{X+}^{(\ell)} : \mathbb{R}^{d_\ell} \times \mathbb{R}^{d_\ell} \times \mathbb{R}^{K-1} \longrightarrow \mathbb{R}^{p_\ell}.$$

For every pure paradigm

$$X \in \{\text{EN}, \text{EE}, \text{WE}\},$$

the update function has the form

$$\phi_X^{(\ell)} : \mathbb{R}^{d_\ell} \times \mathbb{R}^{p_\ell} \longrightarrow \mathbb{R}^{d_{\ell+1}},$$

and produces

$$h_{v,X}^{(\ell+1)} = \phi_X^{(\ell)}(h_v^{(\ell)}, m_{v,X}^{(\ell+1)}).$$

For every Pattern paradigm

$$X+ \in \{\text{EE+}, \text{WE+}\},$$

the update function similarly has the form

$$\phi_{X+}^{(\ell)} : \mathbb{R}^{d_\ell} \times \mathbb{R}^{p_\ell} \longrightarrow \mathbb{R}^{d_{\ell+1}},$$

and produces

$$h_{v,X+}^{(\ell+1)} = \phi_{X+}^{(\ell)}(h_v^{(\ell)}, m_{v,X+}^{(\ell+1)}).$$

A concrete implementation consistent with the EO-Pattern construction is

$$\psi_{X+}^{(\ell)}(h_v^{(\ell)}, h_u^{(\ell)}, \chi_{vu}^X) = \text{MLP}_{\psi, X+}^{(\ell)}(h_v^{(\ell)} \parallel h_u^{(\ell)} \parallel \chi_{vu}^X),$$

and

$$\phi_{X+}^{(\ell)}(h_v^{(\ell)}, m_{v,X+}^{(\ell+1)}) = \text{MLP}_{\phi, X+}^{(\ell)}(h_v^{(\ell)} \parallel m_{v,X+}^{(\ell+1)}),$$

where $\parallel$ denotes concatenation. For the pure layers, the descriptor is omitted:

$$\psi_X^{(\ell)}(h_v^{(\ell)}, h_u^{(\ell)}) = \text{MLP}_{\psi, X}^{(\ell)}(h_v^{(\ell)} \parallel h_u^{(\ell)}).$$

9. Experimental evaluation

Code and experiments are available at:

<https://github.com/adriruiz12/hypergraph-random-walk-encodings>.

The submitted thesis validated the EE paradigm with three experiments: a synthetic twin task, a co-citation benchmark, and a P/Q/R complementarity gadget. This section extends that evaluation to the full EN/EE/WE family, run under a single shared protocol.

One exception is deliberate. The node-classification experiments select the checkpoint by validation accuracy alone. The graph-classification twin task instead breaks ties on validation loss. The reason is that accuracy on the twin task saturates almost immediately, since the task is separable; selecting on accuracy alone would then freeze whichever model happens to reach the maximum first, even if it is still under-converged.

Rather than repeating each thesis experiment three times, once per walk, two of the five experiments are organized as mirror pairs, each isolating one axis of hypergraph structure: cardinality separation (Section 9.1) against multiplicity separation (Section 9.2), and complementarity relative to EE (Section 9.4) against complementarity relative to WE (Section 9.5). The fifth experiment, the Cora/Citeseer benchmark, stands on its own as a test on real data. Two experiments are new (multiplicity separation, and complementarity relative to the WE factorization); the remaining three recover the thesis experiments, now run with the extended family of layers.

All code and result files accompany this note. Alongside the eight variants, every node-classification experiment also reports three baselines: a feature-only MLP, clique-expansion GCN and GIN, and *AllDeepSets* (*mean*), an in-house Deep Sets-style incidence baseline in the spirit of Chien et al. (2022) that uses mean rather than sum aggregation in both propagation directions.

The graph-classification twin task of Section 9.1 reports only the two clique-expansion baselines. Its vertex features are constant, so a feature-only MLP is at chance by construction. And AllDeepSets reads exactly the incidence structure the twins are built to differ in, which is not the pairwise control the task calls for.

Every experiment runs the same eight-variant ladder. Variants (A)-(E) are exactly those of the submitted thesis, with (A) reinterpreted as the pure EN layer of Equation (3) and (D) as the EO-Pattern model of Equation (7). Variants (F)-(H) are their WE counterparts, mirroring (B), (D), and (E); the ladder has no WE counterpart of (C), so the WE side isolates χ^{WE} jointly with P^{WE} but not on its own, and the super-additivity claimed for WE below rests only on comparing (F), (G), and (H).

Table 2: the eight-variant ablation ladder shared by all experiments. (A), (B) and (F) use a kernel alone; (C) isolates the descriptor from its kernel by attaching an EE-induced descriptor to an EN-weighted layer, and is not a pure paradigm; (D) and (G) combine both ingredients, with (E) and (H) as their negative controls.

Variant	Kernel	Descriptor	Role
(A)	P^{EN}	—	pure EN layer, Eq. (3)
(B)	P^{EE}	—	pure EE layer, Eq. (5)
(C)	P^{EN}	χ^{EE}	descriptor isolated from the EE kernel
(D)	P^{EE}	χ^{EE}	EE-Pattern (EO-Pattern), Eq. (7)
(E)	P^{EE}	shuffled χ^{EE}	negative control for (D)
(F)	P^{WE}	—	pure WE layer, Eq. (11)
(G)	P^{WE}	χ^{WE}	WE-Pattern, Eq. (13)
(H)	P^{WE}	shuffled χ^{WE}	negative control for (G)

All layers share the message and update architecture of Section 8: they differ only in the scalar aggregation weight $P^X(v, u)$ and in whether a descriptor is concatenated inside ψ . Since all three kernels are row-stochastic (Proposition 5.1), the aggregation modes are on a common scale and the comparison is not confounded by message magnitude. Two structural identities, both proved below, make two of the experiments diagnostic by construction and organize the reading of the rest.

Three of the experiments (Sections 9.2, 9.4 and 9.5) are node-classification gadgets built by the same protocol, stated here once. Each is a single hypergraph formed by 450 disjoint copies of a fixed local pattern, generated once from a fixed data seed, so that only the split and the weight initialization vary across seeds. Every copy contains one target vertex v , whose label is to be predicted and whose own feature is neutral; one informative neighbor, carrying the true label of v ; and m decoy neighbors of each decoy type in the gadget (one type in the multiplicity gadget, two in each complementarity gadget), carrying independently drawn random classes. A *filler* is a fresh neutral vertex introduced only to bring a hyperedge up to its intended cardinality. Fillers are never reused, so each hyperedge receives its own, and they are ordinary neighbors of v with a pair signature of their own. Features are one-hot over the two classes, with neutral vertices at zero, plus additive Gaussian noise of standard deviation 0.01. The split is class-stratified 50%/25%/25% over the target vertices only, so the majority-class rate over the targets, slightly under 55%, is the floor that a structure-blind model attains. Figure 3 draws the neighbor types of the three gadgets.

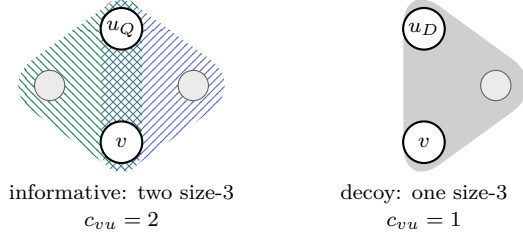
Proposition 9.1 (Coincidence on uniform hypergraphs). *If H is r -uniform, then for every adjacent ordered pair (v, u) ,*

$$P^{\text{EE}}(v, u) = P^{\text{WE}}(v, u), \quad \chi_{vu}^{\text{EE}} = \chi_{vu}^{\text{WE}} = \mathbf{c}_r.$$

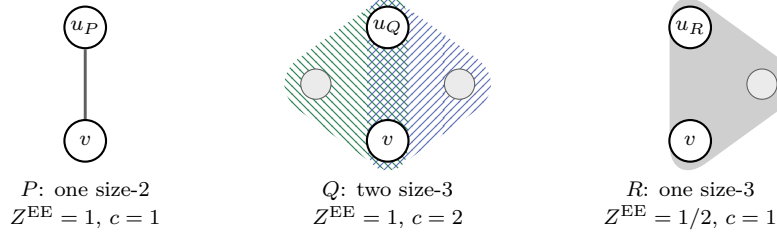
Proof. If $|e| = r$ for every hyperedge, then $Z_{vu}^{\text{EE}} = \sum_{e \in \mathcal{E}(v, u)} (r-1)^{-1} = c_{vu}/(r-1)$ and $s_H(v) = \sum_{e \in \mathcal{E}(v)} (r-1) = (r-1)d_H(v)$. Hence

$$P^{\text{EE}}(v, u) = \frac{Z_{vu}^{\text{EE}}}{d_H(v)} = \frac{c_{vu}}{(r-1)d_H(v)} = \frac{c_{vu}}{s_H(v)} = P^{\text{WE}}(v, u).$$

Multiplicity gadget



EE gadget



WE gadget

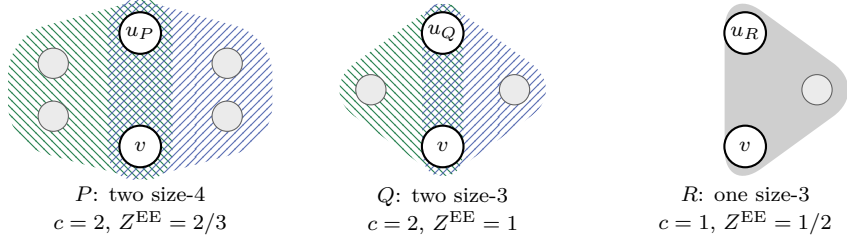


Figure 3: neighbor types of the three gadgets, drawn from the target vertex v . Small grey discs are fillers. When a neighbor is reached through two hyperedges these are *distinct* hyperedges, hatched in opposite directions here, and each carries its own fillers; that is what separates $c_{vu} = 2$ from a single larger hyperedge. Each gadget attaches one informative neighbor and m decoys of each decoy type to every target.

Every shared hyperedge has cardinality r , so both histograms place all their mass in bin $b(r)$, giving $\chi_{vu}^{EE} = \chi_{vu}^{WE} = \mathbf{c}_r$. $\square$

Proposition 9.2 (Pairwise information equivalence of the two Pattern layers). *For an adjacent ordered pair (v, u) , let $n_r(v, u) := |\{e \in \mathcal{E}(v, u) : |e| = r\}|$ be the number of shared hyperedges of each cardinality. The maps*

$$(Z_{vu}^{EE}, \chi_{vu}^{EE}) \longleftrightarrow (n_r(v, u))_{r \geq 2} \longleftrightarrow (c_{vu}, \chi_{vu}^{WE})$$

are mutually invertible under the canonical resolution $K := \max_e |e|$ of Section 3.1 (or any $K \geq \max_e |e|$), since no cardinality then exceeds K and every bin stays unambiguous. Consequently the EE-Pattern and WE-Pattern layers receive, at the pair level, the same information; they differ only in the vertex normalization ($d_H(v)$ versus $s_H(v)$) and in how that information is split between the aggregation weight and the message feature.

Proof. With exact resolution, the nonzero entries of $Z_{vu}^{\text{EE}} \chi_{vu}^{\text{EE}}$ are exactly $\{n_r(v, u)/(r-1)\}_r$, from which each $n_r(v, u)$ is recovered by multiplying bin $b(r)$ by $r-1$; conversely $(n_r)_r$ determines $Z_{vu}^{\text{EE}} = \sum_r n_r/(r-1)$ and its normalized histogram. Likewise the nonzero entries of $c_{vu} \chi_{vu}^{\text{WE}}$ are exactly $\{n_r(v, u)\}_r$, so $(c_{vu}, \chi_{vu}^{\text{WE}})$ and $(n_r)_r$ determine each other. Composing the two bijections gives the claim on $(Z_{vu}^{\text{EE}}, \chi_{vu}^{\text{EE}}) \leftrightarrow (c_{vu}, \chi_{vu}^{\text{WE}})$. The enriched layers receive $P^{\text{EE}}(v, u) = Z_{vu}^{\text{EE}}/d_H(v)$ rather than Z_{vu}^{EE} itself, but $d_H(v)$ is a single factor shared by every neighbor $u \in N_H(v)$, and it is recovered from the whole row by row-stochasticity: $\sum_{u \in N_H(v)} Z_{vu}^{\text{EE}} = d_H(v)$ once each Z_{vu}^{EE} is reconstructed from $(P^{\text{EE}}(v, u), \chi_{vu}^{\text{EE}})$ up to that common factor. The same argument recovers $s_H(v)$ on the WE side. Hence the bijection descends to what the enriched layers actually receive: the row $(P^{\text{EE}}(v, u), \chi_{vu}^{\text{EE}})_{u \in N_H(v)}$ and the row $(P^{\text{WE}}(v, u), \chi_{vu}^{\text{WE}})_{u \in N_H(v)}$ determine each other. $\square$

Proposition 9.2 is the reason complementarity is a property of the chosen factorization rather than of the hypergraph: the same pair information $(n_r)_r$ is repackaged differently by each paradigm, and a construction built against one packaging need not be balanced against the other. Experiments 9.4 and 9.5 demonstrate this in both directions.

9.1. Cardinality separation on twin hypergraphs

This is the synthetic twin task of the submitted thesis, run with the full ladder. Each example is a pair of hypergraphs on a shared vertex set with shared, constant vertex features: one member uses a set of cliques as its hyperedges, the other replaces every clique by its two-element subsets. Both members have identical clique expansions (Figure 4), so any function of the pairwise graph alone receives identical input for the two classes and is bounded at chance. We use 300 base graphs, each yielding one twin pair, split by base graph so that both twins always land in the same split, two layers, width 64, and ten seeds.

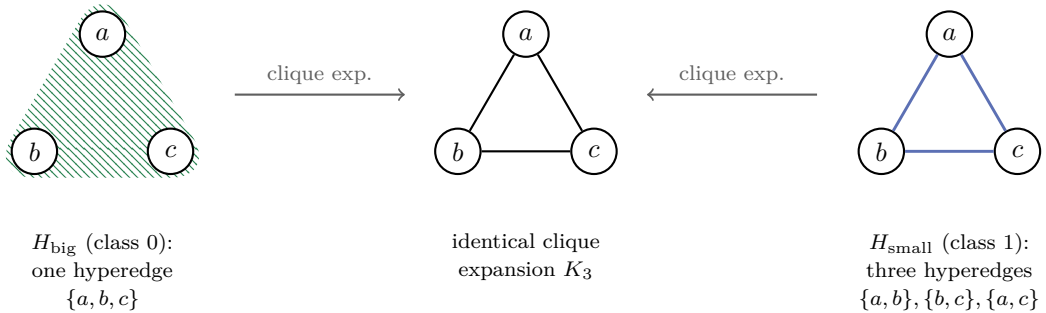


Figure 4: one twin pair. The two members share their vertex set and their (constant) vertex features, and induce the same clique expansion, so every function of the pairwise graph alone receives identical input for the two classes. They differ only in hyperedge cardinality, which is what χ^{EE} and χ^{WE} expose: a pair inside the size-3 hyperedge gets $\mathbf{c}_3$ on the left and $\mathbf{c}_2$ on the right.

Table 3: twin hypergraph classification (mean $\pm$ std over ten seeds). Best results in bold.

Model	Accuracy (%)	Macro-F1 (%)
Clique-GCN (Kipf and Welling, 2017)	50.0 $\pm$ 0.0	33.3 $\pm$ 0.0
Clique-GIN (Xu et al., 2019)	50.0 $\pm$ 0.0	38.1 $\pm$ 7.0
(A) EN	50.0 $\pm$ 0.0	33.3 $\pm$ 0.0
(B) EE	50.0 $\pm$ 0.0	33.3 $\pm$ 0.0
(C) EN + χ^{EE}	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0
(D) EE-Pattern	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0
(E) EE-Pattern, shuffled χ^{EE}	50.0 $\pm$ 0.0	33.3 $\pm$ 0.0
(F) WE	50.0 $\pm$ 0.0	33.3 $\pm$ 0.0
(G) WE-Pattern	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0
(H) WE-Pattern, shuffled χ^{WE}	50.0 $\pm$ 0.0	33.3 $\pm$ 0.0

Two observations follow. First, all three pure kernels sit at chance, for the same reason and not because they coincide: the twins do differ in P^{EE} and P^{WE} , but the vertex features are constant, and a row-stochastic weight maps a constant input to that same constant, so no aggregation weight can break the symmetry. Only a quantity entering ψ as a feature can. Second, the WE descriptor separates the twins exactly as the EE descriptor does: for a pair contained in a single clique of size s , $\chi_{vu}^{\text{WE}} = \mathbf{c}_s$ in the clique hypergraph and $\mathbf{c}_2$ in the decomposed one, so the two classes receive disjoint descriptor values. The controls (E) and (H) return to chance.

9.2. Multiplicity separation

This experiment is the mirror image of Section 9.1: it varies co-occurrence multiplicity at fixed cardinality, where the previous experiment varied cardinality at fixed multiplicity. It is a node-classification gadget in which every hyperedge has size three, so the hypergraph is 3-uniform. For each target vertex v , one informative neighbor u_Q is connected through two size-3 hyperedges $\{v, u_Q, x_1\}$ and $\{v, u_Q, x_2\}$, so that $c_{vu_Q} = 2$, and carries the true label of v in its feature, while $m = 4$ decoys are each connected through one size-3 hyperedge $\{v, u_D, x\}$, so that $c_{vu_D} = 1$, and carry random classes. Every hyperedge receives its own filler, so $x_1 \neq x_2$ and all fillers are distinct; at $m = 4$ the target therefore has $d_H(v) = 6$ incident hyperedges and 11 neighbors: u_Q , the four decoys, and six fillers. The informative neighbor is thus identifiable only from how many hyperedges it shares with v . We use 450 gadgets, one layer, width 64, and ten seeds.

By Proposition 9.1, 3-uniformity forces $\chi_{vu}^{\text{EE}} = \chi_{vu}^{\text{WE}} = \mathbf{c}_3$ for every adjacent pair, so both descriptors are constant and provably carry no information, and it forces $P^{\text{EE}} = P^{\text{WE}}$ identically. Consequently variants (B) and (F) must return the same number, as must (D), (E), (G), and (H): under 3-uniformity these are literally the same model, so the agreement is a built-in correctness check, not a finding.

Table 4: multiplicity gadget (mean $\pm$ std over ten seeds; 450 gadgets, $m = 4$, width 64, one layer). The last column is the gain over the pure EN layer (A). Best results in bold.

Model	Accuracy (%)	Δ over (A)
MLP (features only)	54.0 ± 0.0	
Clique-GCN	70.0 ± 3.8	
Clique-GIN	70.4 ± 3.5	
AllDeepSets (mean)	54.1 ± 0.3	
(A) EN	70.4 ± 2.7	—
(B) EE	86.8 ± 2.2	+16.4
(C) EN + χ^{EE}	69.6 ± 4.4	−0.9
(D) EE-Pattern	85.2 ± 4.2	+14.8
(E) EE-Pattern, shuffled χ^{EE}	85.2 ± 4.2	+14.8
(F) WE	86.8 ± 2.2	+16.4
(G) WE-Pattern	85.2 ± 4.2	+14.8
(H) WE-Pattern, shuffled χ^{WE}	85.2 ± 4.2	+14.8

The result is the exact mirror of Table 3: the kernels separate the classes and no descriptor can. The predicted equalities (B) = (F) and (D) = (E) = (G) = (H) hold to the reported precision, as Proposition 9.1 requires. The near-equality (C) $\approx$ (A) is the descriptor-inertness of Proposition 9.1 observed in practice: a constant descriptor adds nothing, and shuffling a constant is the identity, which is why the controls (E) and (H) do not fall.

9.3. Co-citation node classification

Node classification on the Cora and Citeseer co-citation hypergraphs (Yadati et al., 2019), with class-stratified 50%/25%/25% splits generated independently for this note, following the split proportions of Chien et al. (2022), two layers, width 128, twenty seeds. Incidence matrices are built with SciPy and cross-checked against TopoNetX (Hajij et al., 2024) when available. The baselines are those listed at the start of Section 9, as in the thesis.

Table 5: node classification accuracy (%) on the Cora co-citation hypergraph (mean $\pm$ std over twenty seeds). *all*: all test nodes; *active*: nodes with $d_H(v) > 0$.

Model	all	active
MLP (features only)	73.8 ± 1.4	75.1 ± 2.0
Clique-GCN	79.9 ± 1.5	85.0 ± 1.7
Clique-GIN	76.6 ± 1.3	82.8 ± 1.5
AllDeepSets (mean)	75.2 ± 1.5	76.5 ± 1.9
(A) EN	76.3 ± 1.2	80.4 ± 1.7
(B) EE	76.5 ± 1.2	80.8 ± 1.5
(C) EN + χ^{EE}	76.5 ± 1.2	80.7 ± 1.7
(D) EE-Pattern	76.2 ± 1.2	80.5 ± 1.8
(E) EE-Pattern, shuffled χ^{EE}	76.3 ± 1.3	80.5 ± 1.8
(F) WE	76.4 ± 1.0	80.7 ± 1.5
(G) WE-Pattern	76.2 ± 1.0	80.4 ± 1.6
(H) WE-Pattern, shuffled χ^{WE}	76.1 ± 1.3	80.2 ± 1.6

Table 6: node classification accuracy (%) on the Citeseer co-citation hypergraph (mean $\pm$ std over twenty seeds). *all*: all test nodes; *active*: nodes with $d_H(v) > 0$.

Model	all	active
MLP (features only)	72.2 ± 0.9	74.9 ± 1.9
Clique-GCN	73.2 ± 1.4	76.3 ± 2.1
Clique-GIN	71.2 ± 1.0	73.9 ± 2.2
AllDeepSets (mean)	73.1 ± 1.0	76.0 ± 1.8
(A) EN	72.4 ± 1.0	75.7 ± 1.7
(B) EE	72.5 ± 1.0	75.9 ± 1.5
(C) EN + χ^{EE}	72.5 ± 1.1	75.7 ± 2.1
(D) EE-Pattern	72.7 ± 1.0	75.8 ± 2.1
(E) EE-Pattern, shuffled χ^{EE}	72.6 ± 1.0	75.7 ± 2.2
(F) WE	72.5 ± 1.0	75.9 ± 1.6
(G) WE-Pattern	72.5 ± 1.0	75.7 ± 2.0
(H) WE-Pattern, shuffled χ^{WE}	72.6 ± 1.0	75.8 ± 1.9

The submitted thesis already reported a negative result for the EE paradigm here, and the ladder (A)–(E) reproduces it on both datasets: the variants lie within one standard deviation of each other, the shuffled control (E) matches the full model (D), and a residual Clique-GCN is stronger on Cora. The WE rows answer whether that flatness is specific to the equal-edges quantities or a property of the data. The relevant structural fact is that co-citation hyperedges overlap in few vertices, so $c_{vu} = 1$ for about four fifths of adjacent ordered pairs on both datasets (80.9% on Cora, 80.0% on Citeseer). On such a pair the unique shared hyperedge gives $\chi_{vu}^{\text{WE}} = \mathbf{c}_{|e|} = \chi_{vu}^{\text{EE}}$, so the two descriptors agree identically there, and hence on 85.9% (Cora) and 83.5% (Citeseer) of all adjacent pairs. The WE kernel degenerates at the level of vertices rather than pairs: $P^{\text{WE}}(v, \cdot) = P^{\text{EN}}(v, \cdot)$ exactly when c_{vu} is *constant* over $u \in N_H(v)$, since then $s_H(v) = c|N_H(v)|$ and the common factor c cancels; this holds for 59.8% of the active vertices of Cora and 71.0% of Citeseer (the stricter condition $c_{vu} = 1$ throughout $N_H(v)$ accounts for 58.4% and 67.4%). The WE ladder is indeed flat on both datasets, confirming the second reading: the co-citation regime is degenerate for all three walks rather than unfavourable to one of them. Citeseer, with a wider range of hyperedge cardinalities than Cora ($K = 26$ versus $K = 5$), is if anything flatter still (a 0.29-point range across all eight variants, against 0.46 on Cora), so the flatness is not an artifact of a coarse cardinality resolution.

9.4. Complementarity relative to the EE factorization

This is the P/Q/R gadget of the submitted thesis, run with the full ladder. Each target vertex v has one informative neighbor Q connected through two size-3 hyperedges $\{v, u_Q, x_1\}$ and $\{v, u_Q, x_2\}$, m decoys of type P connected through one size-2 hyperedge $\{v, u_P\}$, and m decoys of type R connected through one size-3 hyperedge $\{v, u_R, x\}$. Every size-3 hyperedge receives its own filler, so at $m = 4$ the target has $d_H(v) = 10$ incident hyperedges and 15 neighbors: Q , four P , four R , and six fillers, the last two groups sharing one pair signature. Writing $c_{vu} = |\mathcal{E}(v, u)|$, the neighbor types have the signatures in Table 7.

Table 7: signatures of the three neighbor types in the EE gadget. χ^{EE} collapses Q with R ; P^{EE} (through Z^{EE}) collapses Q with P ; only the pair $(Z^{\text{EE}}, \chi^{\text{EE}})$ singles out Q .

Type	shared $ e $	Z_{vu}^{EE}	c_{vu}
P	2	1	1
Q	3	1	2
R	3	1/2	1

Table 8: EE complementarity gadget (mean $\pm$ std over ten seeds; 450 gadgets, $m = 4$, width 64, one layer). Best result in bold.

Model	Accuracy (%)	Δ over (A)
MLP (features only)	54.2 ± 2.1	
Clique-GCN	63.5 ± 3.8	
Clique-GIN	63.3 ± 3.5	
AllDeepSets (mean)	54.9 ± 0.0	
(A) EN	62.7 ± 4.0	—
(B) EE	68.1 ± 2.9	+5.5
(C) EN + χ^{EE}	61.1 ± 3.6	−1.6
(D) EE-Pattern	80.1 ± 3.1	+17.4
(E) EE-Pattern, shuffled χ^{EE}	68.1 ± 3.2	+5.4
(F) WE	75.8 ± 3.1	+13.1
(G) WE-Pattern	80.0 ± 2.5	+17.3
(H) WE-Pattern, shuffled χ^{WE}	75.7 ± 2.8	+13.0

The EE block reproduces the thesis result, though χ^{EE} in isolation is weaker than the thesis reported: P^{EE} alone gives a modest gain (+5.5), while χ^{EE} alone, attached to the EN kernel, is not merely unhelpful but slightly harmful (−1.6) relative to EN on its own. Neither ingredient is individually strong, and their combination is (+17.4), essentially the same gain achieved by (G), which matches (D) almost exactly (80.0 versus 80.1), in this particular gadget. Proposition 9.2 guarantees that (D) and (G) are trained on the same pair information here, not that two independently trained networks converge to the same accuracy; the close numerical match is consistent with that guarantee but is not implied by it, and the mirror gadget of Section 9.5 shows the two need not coincide (D and G differ by 8.3 points at $m = 4$ and G is significantly stronger than D at $m = 2$). Permuting χ^{EE} returns the model to the kernel-only level. The new information is the WE block. The gadget is not neutral with respect to WE: by Table 7, Q is the only neighbor with $c_{vu} = 2$, and $P^{\text{WE}} = c_{vu}/s_H(v)$ is exactly the kernel that retains multiplicity. Hence the pure WE layer (F) reaches +13.1 with no descriptor at all: the discrimination that the EE paradigm can perform only jointly, the WE kernel performs on its own.

9.5. Complementarity relative to the WE factorization

The mirror construction targets the WE factorization. Each target has one informative neighbor Q connected through two size-3 hyperedges $\{v, u_Q, x_1\}$ and $\{v, u_Q, x_2\}$, m decoys of type P each connected through two size-4 hyperedges $\{v, u_P, x, x'\}$ and $\{v, u_P, x'', x'''\}$, and m decoys of type R connected through one size-3 hyperedge $\{v, u_R, x\}$. Again every hyperedge receives its own fillers, so at $m = 4$ the target has $d_H(v) = 14$ incident hyperedges and 31 neighbors: Q , four P ,

four R , and 22 fillers. The signatures are shown in Table 9: now P^{WE} (through c_{vu}) collapses Q with P , χ^{WE} collapses Q with R , and only the pair singles out Q . Symmetrically, the EE masses Z^{EE} are all distinct, so P^{EE} alone identifies Q .

Table 9: signatures of the three neighbor types in the WE gadget. The pair $(c_{vu}, \chi^{\text{WE}})$ singles out Q ; each alone collapses it with a decoy. Because Z^{EE} is distinct for all three types, P^{EE} alone identifies Q . The neutral fillers form a fourth signature, $(|e| = 4, c_{vu} = 1, Z^{\text{EE}} = 1/3)$ for those of the P hyperedges and the signature of R for the rest; both separations above survive their presence.

Type	shared $ e $	c_{vu}	Z_{vu}^{EE}
P	4	2	2/3
Q	3	2	1
R	3	1	1/2

The WE gadget attaches two size-4 hyperedges per P -decoy, so at $m = 4$ the informative neighbor carries only $P^{\text{WE}}(v, u_Q) = 2/36 \approx 5.6\%$ of the aggregation mass, against the corresponding EE share $P^{\text{EE}}(v, u_Q) = 1/10 = 10\%$ in the EE gadget of Section 9.4 (each gadget measured under the kernel it was built against). We therefore also report a dilution-matched control with $m = 2$, for which the share is $2/20 = 10\%$; the setting is fixed by that computation, not chosen from the outcome.

Table 10: WE complementarity gadget (mean $\pm$ std over ten seeds; 450 gadgets, width 64, one layer). Left: as specified, $m = 4$ decoys per type. Right: dilution-matched control, $m = 2$. Best result in each column in bold.

Model	$m = 4$	$m = 2$ (matched)
MLP (features only)	54.4 ± 1.3	54.0 ± 0.0
Clique-GCN	65.2 ± 2.7	75.3 ± 4.7
Clique-GIN	62.7 ± 4.3	69.6 ± 3.8
AllDeepSets (mean)	54.9 ± 0.0	54.0 ± 0.0
(A) EN	55.9 ± 3.2	70.3 ± 3.4
(B) EE	72.7 ± 7.1	85.6 ± 4.1
(C) EN + χ^{EE}	54.9 ± 0.0	73.7 ± 7.1
(D) EE-Pattern	73.6 ± 10.4	87.7 ± 2.3
(E) EE-Pattern, shuffled χ^{EE}	68.2 ± 9.4	86.5 ± 2.0
(F) WE	63.8 ± 6.5	74.9 ± 2.8
(G) WE-Pattern	65.3 ± 12.9	90.9 ± 1.5
(H) WE-Pattern, shuffled χ^{WE}	60.5 ± 7.0	72.4 ± 6.8

The mirror prediction on the EE side is confirmed in both columns: P^{EE} alone (B) is the strongest single-ingredient variant, exactly as P^{WE} alone was in the EE gadget. At $m = 4$, (C) collapses to the majority-class predictor (54.9 ± 0.0), a point below (A) and well inside its standard deviation of 3.2, because χ^{EE} confuses Q with R here just as EN does. On the WE side, the intended super-additivity is present at both dilutions in direction, but only the matched control resolves it cleanly: at $m = 2$, the pure WE layer (F) adds +4.6 over (A), its shuffled control (H) falls further still to +2.1 rather than merely matching (F), and the WE-Pattern layer (G) reaches +20.6, the same shape of super-additivity as the +17.4 of the

EE gadget and, by a paired test across seeds, significantly above both (F) and (H) ($p < 0.005$), the clearest evidence in either gadget that the WE factorization is not redundant with the EE one. At $m = 4$ the (G) – (F) gap of +1.5 sits well inside a standard deviation of 12.9 and is not established; the large variance is itself the signature of dilution. Symmetrically, (E) $\approx$ (D) at $m = 2$ (86.5 versus 87.7): given P^{EE} , the descriptor χ^{EE} is close to inert here, because Z^{EE} already identifies Q .

9.6. Summary

The five experiments support three claims. First, cardinality and multiplicity are independent axes of hypergraph structure, and Sections 9.1 and 9.2 isolate one each, with the other axis neutralized by construction rather than by tuning: descriptors separate where kernels provably cannot, and kernels separate where descriptors provably cannot. Second, by Proposition 9.2, the two Pattern layers receive the same pair information and differ only in how it is split between weight and feature. Third, and consequently, complementarity is a property of the chosen factorization, not of the hypergraph information, and Sections 9.4 and 9.5 exhibit it in both directions: each gadget needs both ingredients of the paradigm it was built against, while the other paradigm resolves the same task with its kernel alone. On natural co-citation data (Section 9.3), all three walks appear to fall in a degenerate regime where these distinctions collapse. EE and WE therefore do not carry different information about a pair of vertices; they carry different factorizations of the same information, and which factorization suits a task is a property of the task. EN is not part of this equivalence: it discards both cardinality and multiplicity, retaining only whether the pair is adjacent at all.

10. Current scope

This note provides a mathematical formulation and conceptual comparison of the EN, EE, and WE message-passing paradigms, and evaluates all five resulting layers under a common protocol (Section 9) on synthetic diagnostics and on the Cora and Citeseer co-citation benchmarks. The EN layer is identified with the uniform aggregation baseline of the submitted work, and the EE-Pattern layer with the submitted EO-Pattern model; the WE and WE-Pattern layers, together with the multiplicity and WE-complementarity experiments, are new. The experiments support the reading that EE and WE encode different factorizations of the same pairwise information (Proposition 9.2) rather than different information, while EN discards it outright; the usefulness of each paradigm is task-dependent in a way predictable from what each kernel compresses or discards. Two directions remain, in parallel. One is architectural: moving from these controlled ablation layers to using the descriptors as edge encodings inside a strong existing hypergraph backbone, of the $V \rightarrow \mathcal{E} \rightarrow V$ kind sketched in Section 7, where the comparison of interest is that backbone with and without the random-walk information. The other is theoretical: reviewing the relevant results in the hypergraph neural network literature to assess whether analogues can be established for the EN/EE/WE construction presented here.

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A. Relation to the relational message-passing framework

The relational message-passing framework of Taha et al. (2025) describes propagation through a set of entities, a collection of relations, their associated shift operators, message functions, and an update function. The pure EN, EE, and WE layers can be represented as particular cases of this framework. This appendix formalizes that correspondence; it is not used elsewhere in this note, since the theoretical questions currently pursued for this construction do not go through relational structures.

Take the entity set to be

$$S = V.$$

Define the binary adjacency relation

$$R_{\text{adj}} := \{(v, u) \in V \times V : u \in N_H(v)\}.$$

For each paradigm

$$X \in \{\text{EN}, \text{EE}, \text{WE}\},$$

define the corresponding shift operator by

$$A_{v,u}^{R_{\text{adj}},X} := P^X(v, u).$$

Since R_{adj} is binary, its aggregated influence matrix coincides with its shift operator:

$$\tilde{A}^X = A^{R_{\text{adj}},X} = P^X.$$

The associated influence graph has a directed edge

$$u \longrightarrow v$$

whenever $P^X(v, u) > 0$, with influence weight

$$w_{u \rightarrow v} = P^X(v, u).$$

The direction is from u to v because the representation of u contributes to the update of v . Although the original hypergraph is undirected, the influence weights need not be symmetric because the transition kernels are normalized with respect to the target vertex.

To represent the explicit dependence of the update function on the current representation $h_v^{(\ell)}$, we also introduce the unary identity relation

$$R_{\text{id}} := \{(v) : v \in V\},$$

with shift operator

$$A_v^{R_{\text{id}}} := 1$$

and message function

$$\psi_{\text{id}}^{(\ell)}(h_v^{(\ell)}) = h_v^{(\ell)}.$$

The update function then combines the identity message with the message received through R_{adj} :

$$h_{v,X}^{(\ell+1)} = \phi_X^{(\ell)}\left(\psi_{\text{id}}^{(\ell)}(h_v^{(\ell)}), m_{v,X}^{(\ell+1)}\right).$$

Since R_{id} is unary, it supplies the current representation of the target vertex to the update function but does not represent influence from a different entity. Therefore, it does not add an edge to the influence graph induced by P^X .

Under this interpretation, EN, EE, and WE use the same relational support but different shift operators. The relational framework therefore provides a general language for representing the three pure random-walk-induced layers, while the present construction specifies concrete shift operators derived from the incidence structure of a hypergraph.

The Pattern variants additionally use the pairwise descriptors

$$\chi_{vu}^{\text{EE}} \quad \text{and} \quad \chi_{vu}^{\text{WE}}.$$

These descriptors are not contained in the scalar shift operators $P^X(v, u)$, but they can still be represented within the relational message-passing framework by introducing an auxiliary entity p_{vu} for every ordered adjacent pair, with fixed feature

$$h_{p_{vu}} = \chi_{vu}^X.$$

The corresponding relation is ternary:

$$R_{\text{Pattern}}^X := \{(v, u, p_{vu}) : u \in N_H(v)\},$$

with shift operator

$$A_{v,u,p_{vu}}^{R_{\text{Pattern}}^X} = P^X(v, u).$$

Its message function receives the target representation, the source representation, and the pairwise descriptor:

$$\psi_{\text{Pattern}}^{X,(\ell)}(h_v^{(\ell)}, h_u^{(\ell)}, h_{p_{vu}}) = \psi_{X+}^{(\ell)}(h_v^{(\ell)}, h_u^{(\ell)}, \chi_{vu}^X).$$

Therefore, both the pure and Pattern variants can be represented within the relational framework. The pure variants use a binary relation, whereas the Pattern variants use an auxiliary pair entity and a ternary relation.